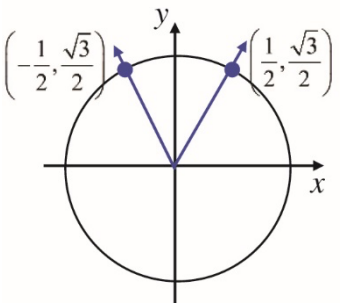
Solve Trigonometric Equations in Linear Form

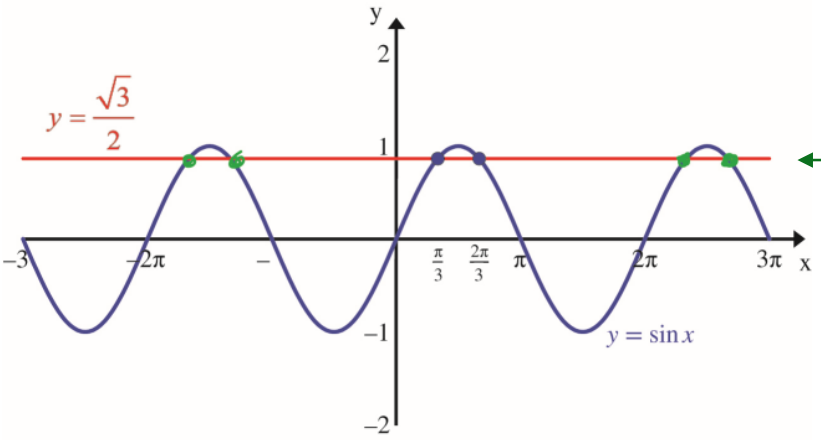
Example: Solve. $\sin x = \frac{\sqrt{3}}{2}$

From the unit circle, we can see that there are 2 values of x between 0 and 2π that will solve the equation: $x = \frac{\pi}{3}$ and $x = \frac{2\pi}{3}$.



However, there are actually infinitely many solutions to the equation:

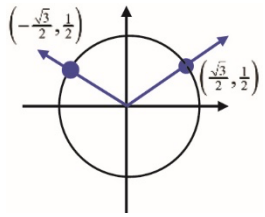
$x = \frac{\pi}{3} + 2n\pi$ and $x = \frac{2\pi}{3} + 2n\pi$ (where n is any integer).



all intersections are solutions

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1. Solve the equation on the interval $[0, 2\pi)$. $3\csc x = 4 + \csc x$



2. Solve the equation over the set of real numbers. $\sqrt{2}\cos x + 5 = 4$

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Solve Trigonometric Equations Involving Multiple or Compound Angles

Solve the equation on the interval $[0, 2\pi)$.

3. $8\cot 2x - 5 = 3$

   Isolate $\cot 2x$.

   Give the general solution from the unit circle.

   To find solutions on the interval $[0, 2\pi)$, substitute $n = \ldots, -1, 0, 1, 2, \ldots$

4. $\cos\frac{x}{2} = \frac{\sqrt{3}}{2}$

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Solve Higher-Degree Trigonometric Equations

Solve the equation on the interval $[0, 2\pi)$.

5. $2\sin^2 x + 5\sin x + 3 = 0$

Use Identities to Solve Trigonometric Equations

Solve on the interval $[0, 2\pi)$.

6. $\sin 2x = \cos x$

7. $\sin x = \cos x$

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Use Inverse Functions to Express Solutions to Trigonometric Equations

8. Solve on the interval $[0, 2\pi)$. $4\cos^2 x - 7\cos x + 3 = 0$

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